

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Code of Conduct

I Beffy A Shanika(192511387)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: beffy

Assessment Tasks

Industry Problem Solving Task

1. An e-commerce company receives clickstream, transaction, and review data from millions of users daily. Propose a big data analytics pipeline and explain how the 5Vs influence architecture design.
2. A healthcare organization collects structured patient records, semi-structured insurance claims, and unstructured medical images. Design an approach to manage and secure this data ecosystem.
3. A telecom company wants to detect churn patterns using high-volume customer interaction data. Identify the big data challenges and recommend suitable analytics and integration strategies.
4. A smart city project generates sensor streams from traffic, air quality, and surveillance systems. Explain how distributed file systems and data integration tools can support large-scale analysis.
5. A financial analytics firm must comply with strict privacy requirements while processing high-velocity transaction data. Design a secure big data solution and justify the controls chosen.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

ANSWER:

1. E-commerce Big Data Analytics Pipeline

An e-commerce company receives clickstream data, transaction data, and customer reviews from millions of users every day. A scalable big data pipeline can be designed as follows:

Data Sources → Data Ingestion → Distributed Storage → Data Processing → Analytics → Visualization/Decision Making

Architecture

- Data Sources: Website clicks, mobile-app activity, transactions, customer reviews.

- Data Ingestion: Tools such as Kafka can collect high-speed streaming data.
- Storage: A distributed file system such as HDFS or cloud object storage can store large volumes of structured, semi-structured, and unstructured data.
- Processing: Spark can process data in batch and real time.
- Analytics: Machine learning and statistical techniques can be used for recommendations, customer behavior analysis, and sales prediction.
- Visualization: Dashboards can display sales trends, customer behavior, and product performance.

Influence of 5Vs

5V Influence on Architecture

Volume Requires distributed and scalable storage.

Velocity Requires real-time data ingestion and stream processing.

Variety Requires systems capable of handling tables, JSON, text, images, and reviews.

Veracity Requires data cleaning, validation, and quality checks.

Value Analytics must convert collected data into useful business insights.

Result: A distributed, scalable pipeline allows the company to process millions of records efficiently and provide personalized recommendations and better business decisions.

2. Healthcare Data Management and Security

The healthcare organization has three major types of data: structured patient records, semi-structured insurance claims, and unstructured medical images

Proposed Approach

Patient Records + Insurance Claims + Medical Images



Data Integration Layer



Secure Data Lake / Distributed Storage



Data Processing & Analytics



Healthcare Applications

Data Management

1. Structured Data

- Store patient records in relational databases.
- Examples include patient ID, age, diagnosis, and laboratory results.

2. Semi-structured Data

- Store insurance claims in formats such as JSON or XML.
- Use data integration tools to transform and standardize the information.

3. Unstructured Data

- Store medical images such as X-rays and MRI scans in scalable object or distributed storage.

Security Measures

- Encryption: Encrypt data during storage and transmission.
- Authentication: Allow only authorized healthcare staff to access systems.
- Role-Based Access Control: Doctors, nurses, administrators, and analysts receive different permissions.
- Data Masking: Hide sensitive patient information when used for analytics.
- Audit Logs: Record who accessed or modified patient information.
- Backup and Recovery: Maintain secure backups to prevent data loss.

Result: The approach provides scalable storage while protecting sensitive patient information and maintaining data availability and integrity.

3. Telecom Customer Churn Analysis

A telecom company generates huge amounts of customer interaction data from calls, messages, internet usage, complaints, recharge history, and customer-service interactions.

Major Big Data Challenges

1. High Volume

- Millions of customer interactions generate massive datasets.
- Distributed storage and processing are required.

2. High Velocity

- Call and network data is generated continuously.
- Real-time or near-real-time processing may be required.

3. Data Variety

- Data comes from databases, logs, applications, customer feedback, and social media.
- Different formats must be integrated.

4. Data Quality

- Duplicate, missing, or inconsistent customer records can affect churn prediction.

5. Privacy and Security

- Customer information must be protected from unauthorized access.

Recommended Analytics Strategy

- Use descriptive analytics to understand existing customer behavior.
- Use predictive analytics to identify customers likely to leave.
- Analyze factors such as:
 - Reduced usage
 - Frequent complaints
 - Recharge frequency
 - Call duration
 - Network problems
 - Service plan changes

Integration Strategy

Customer Data + Call Records + Recharge Data + Complaints



Data Integration/ETL



Distributed Storage



Data Cleaning & Processing



Churn Analytics



High-Risk Customer Identification



Retention Offers

Result: The telecom company can identify customers at risk of leaving and take preventive actions such as personalized offers and improved customer support.

4. Smart City Big Data Analysis

A smart city generates continuous sensor data from traffic systems, air-quality sensors, and surveillance systems. This creates a high-volume and high-velocity data environment.

Architecture

Traffic Sensors + Air Quality Sensors + Surveillance Systems



Data Integration Tools



Data Ingestion



Distributed File System / Data Lake



Distributed Processing



Analytics & Machine Learning



Smart City Dashboard

Role of Distributed File Systems

A distributed file system such as HDFS can:

- Store huge amounts of sensor data across multiple machines.
- Divide large files into blocks and distribute them across nodes.
- Provide fault tolerance through data replication.
- Allow multiple machines to process data simultaneously.
- Easily scale by adding more storage nodes.

Role of Data Integration Tools

Integration tools help:

- Collect data from different sensor systems.
- Convert data into common formats.
- Clean and transform sensor data.
- Combine data from traffic, air quality, and surveillance systems.
- Transfer data to storage and analytics platforms.

Example

Traffic sensors can identify congestion while air-quality sensors measure pollution. Combining both datasets can help city authorities identify areas where traffic congestion is contributing to poor air quality.

Result: Distributed storage and integration tools enable the smart city to process large-scale sensor data efficiently and support real-time decision making.

5. Secure Big Data Solution for Financial Analytics

A financial analytics firm processes high-velocity transaction data while following strict privacy requirements. The solution must provide security, scalability, real-time processing, and regulatory compliance.

Proposed Architecture

Financial Transactions



Secure Data Ingestion



Encrypted Data Lake / Distributed Storage



Stream & Batch Processing



Analytics / Fraud Detection



Secure Reports & Dashboards

Security Controls

1. Encryption

Encrypt sensitive financial data both at rest and in transit to prevent unauthorized access.

2. Authentication

Use strong authentication, preferably multi-factor authentication (MFA), to verify users.

3. Role-Based Access Control

Users should receive only the permissions required for their jobs.

For example:

- Analysts → Access to approved analytical datasets.
- Administrators → System management.
- Auditors → Access to logs and compliance information.

4. Data Masking

Sensitive information such as account numbers and personal details should be masked when full information is not required.

5. Audit Logging

Maintain logs of data access, modifications, and administrative activities. This helps detect suspicious activity and supports compliance audits.

6. Data Anonymization

Remove or transform personally identifiable information when datasets are used for analytics or research.

7. Real-Time Monitoring

Monitor transaction streams for unusual activities and potential fraud.

8. Backup and Disaster Recovery

Maintain secure backups and recovery mechanisms to protect against system failures, ransomware, or data loss.

Justification

Control	Purpose
Encryption	Protects confidential financial data
Authentication	Prevents unauthorized users
Access control	Limits access based on user roles
Masking/Anonymization	Protects personal information
Audit logs	Supports monitoring and compliance
Real-time monitoring	Detects suspicious transactions
Backup & recovery	Prevents permanent data loss

Result: The proposed architecture allows the firm to process high-speed financial transactions while maintaining confidentiality, integrity, availability, privacy, and regulatory compliance.